



Prompting somatic practice performance with AI-facilitated peer-assisted learning: A self-determination theory perspective

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Abstract

Incorporating peer-assisted learning (PAL) into curricula to foster students' feedback skills, reduce their reliance on teachers, and enhance their confidence in learning as well as strengthening connections among peers has been recommended by scholars; however, the ability of students, who are inexperienced peer tutors, to guide others by providing feedback based on individual peers' needs is frequently questioned. To address this issue, this study proposed an artificial intelligence(AI)-supported PAL mode based on the perspective of self-determination theory (SDT). Additionally, an AI-facilitated PAL system was developed to serve as scaffolding to support students while executing PAL tasks. To explore the effectiveness of this learning mode, this study adopted a quasi-experimental design involving two classes of students enrolled in a physical education course at a technological university. One class of 44 students served as the experimental group adopting the AI-PAL mode, while the other class of 46 students served as the control group adopting the conventional (C-PAL) mode. The collected data were analyzed using SPSS for ANCOVA. The results indicated that the AI-PAL mode significantly enhanced students' skills performance, learning motivation, autonomy, competence, and relatedness, compared to the C-PAL mode. Furthermore, the interview results revealed the contributions of this mode to enhancing learning effectiveness, promoting the quality of PAL, and reducing teachers' workload. This study demonstrates the system development, application, and learning benefits of the AI-PAL mode in the physical education context. The findings could be a reference for those who intend to improve students' skills performances using the PAL approach; moreover, it could also be a reference for developing policies to guide teachers to adopt AI in educational settings.

Keywords Peer-assisted learning (PAL) · Peer tutoring · Intelligent tutoring system (ITS) · Adaptive learning · Human activity recognition · Somatic practices · Physical education

1 Introduction

Providing students with opportunities to choose and monitor their own learning, and fostering their autonomy in face-to-face learning is an important issue in the field of education (Chiang, 2025). However, the most commonly used instructional approach in classrooms involves teachers delivering knowledge while students engage in learning by listening, imitating, and performing specific tasks (Coliñir et al., 2022). This approach often places students in a passive role, where they focus on absorbing extensive information from lectures rather than actively engaging with the learning content (Förster et al., 2022). To address this situation, scholars have advocated for the integration of peer-assisted learning (PAL) in teaching, which aims to shift students away from passively receiving information and instead to foster opportunities for collaborative learning among peers (Coliñir et al., 2022; Guraya & Abdalla, 2020; Topping & Ehly, 2001).

PAL is an instructional strategy through which students gain knowledge and skills by supporting one another. For instance, each student may alternately take on the roles of tutor and learner, gaining valuable experience by guiding peers and receiving feedback from them (Topping & Ehly, 2001). This learning process is grounded in Vygotsky's social constructivism (Guraya & Abdalla, 2020; Williams & Reddy, 2016), which emphasizes that knowledge is co-constructed through negotiation, discussion, and adaptation within social interactions among individuals and their environment (Vygotsky, 1978). This aligns with the principles of PAL, providing a theoretical foundation for implementing this instructional strategy.

Previous studies have indicated that PAL has been widely implemented across various educational fields. It is a highly promising instructional strategy that may enhance the quality of conventional physical education (Jenkinson et al., 2014; Ward & Lee, 2005). In addition, numerous studies have reported that PAL positively impacts students' skills performance, competence, and relatedness (Byl & Topping, 2023; Lam et al., 2024). Nshimiyimana and Cartledge (2020) further suggested that well-designed PAL activities could foster students' autonomy, competence, and relatedness. From the perspective of self-determination theory (SDT), fulfilling these three basic psychological needs can enhance students' intrinsic motivation (Ryan & Deci, 2020). Subsequently, students become more engaged and enthusiastic in their learning, thereby supporting their academic growth. The aforementioned studies on PAL primarily analyzed students' subjective feedback through surveys and interviews, whereas fewer studies have objectively verified the direct influences of PAL on student learning outcomes. Although scholars widely agree that PAL is an effective instructional strategy, future research should focus on identifying the factors that determine its effectiveness (Ala et al., 2021).

Williams and Reddy (2016) conducted a literature review, and pointed out that some research findings have confirmed that PAL can objectively enhance student learning outcomes. However, an equal number of studies have indicated that the intervention of PAL does not lead to improvements in student performance. It is worth noting that there are some challenges associated with the implementation of PAL. For instance, students may be limited by their knowledge levels and critical thinking abilities, which could hinder their capacity to provide adequate assistance to peers;

moreover, students' subjective manners could also affect their judgments in providing feedback to peers. This implies that the feedback presented by students during PAL could be inappropriate or even misleading (Roberts et al., 2024). Several studies have reported that students may lack confidence in the peer feedback they receive during PAL activities (Nshimiyimana & Cartledge, 2020; Swallow et al., 2023). Therefore, past research has indicated that most students desire more professional assistance during PAL to ensure the accuracy of the feedback they provide and to help address challenging problems for which no one knows the answer. Additionally, some scholars have pinpointed that inexperienced novices lack the ability and knowledge to guide their peers; when confronted with complex skills, student learning outcomes may be better supported by experienced professionals or computer-assisted learning (Williams & Reddy, 2016). However, relying on teachers to supervise the implementation of PAL to ensure that all peer feedback is accurate poses a significant challenge given the limited teaching staff and resources (Nshimiyimana & Cartledge, 2020).

In light of these challenges, scholars have suggested employing artificial intelligence (AI) technology to provide immediate feedback, thereby enhancing students' competencies of providing quality feedback to peers (Darvishi et al., 2022; Guo et al., 2024; Lai & Tu, 2024; Sembey et al., 2024). For instance, students could be encouraged to reference AI-generated feedback during peer assessments for reflection, self-regulation and monitoring, ultimately enhancing their feedback quality (Darvishi et al., 2022; Guo et al., 2024; Sembey et al., 2024). As indicated by scholars, adopting AI feedback in a complementary way has great potential for enhancing students' feedback quality (Güner & Er, 2025). However, most of the applications of AI-facilitated PAL have relied on discourse analysis (Chen et al., 2022) and were implemented mainly for knowledge delivery (e.g. Darvishi et al., 2022; Guo et al., 2024). Scholars have argued that simple text-based communication may not fully align with the needs of motor skills learning. During the process of learning motor skills, providing students with visual feedback, combined with adaptive guidance and discussion, can support their progress more effectively (Lin et al., 2020).

As a result, this study referenced previous literature and proposed an AI-supported PAL (AI-PAL) mode from the perspective of SDT. An AI-facilitated PAL system was developed to immediately evaluate students' motor skills performance, as well as to provide adaptive visual feedback, score feedback, and test indicators so as to enable students to perform PAL from a professional and objective perspective. To evaluate the effectiveness of this mode, we conducted a teaching experiment in two classes of a physical education course at a technological university. The following research questions were addressed:

- (1) What differences will there be in students' motor skill performance when using the AI-PAL model compared to using the Conventional PAL (C-PAL) model?
- (2) What differences will there be in students' learning motivation when using the AI-PAL model compared to using the C-PAL model?
- (3) What differences will there be in students' basic psychological needs (i.e., autonomy, competence, and relatedness) when using the AI-PAL model compared to using the C-PAL model?
- (4) What effects do the C-PAL mode and AI-PAL mode have on students' learning?

2 Literature review

2.1 PAL

An instructional strategy that enables students to acquire knowledge and skills through social interactions and mutual support is generally referred to as PAL (Byl & Topping, 2023; Topping & Ehly, 2001; Williams & Reddy, 2016). This strategy emphasizes the completion of learning tasks through mutual support, interaction, and collaborative learning among students (Guraya & Abdalla, 2020), which aligns with the principles of social constructivism, advocating that learning occurs in a social context where students assist each other in learning by sharing knowledge (Atwater, 1996; Williams & Reddy, 2016). Such a peer interaction process is also in line with the cognitive learning theory, suggesting that when peers are at a similar level of knowledge, the atmosphere of learning exchange and interaction tends to be more relaxed. This encourages students to talk about their shortcomings and fears in learning, and hence benefits them in solving learning problems (Guraya & Abdalla, 2020; Herrmann-Werner et al., 2017). Numerous studies have suggested that social constructivism as well as cognitive learning theory can be employed to explain and substantiate the advantages of PAL for student learning (Guraya & Abdalla, 2020; Williams & Reddy, 2016).

PAL consists of various forms, including peer tutoring, peer modeling, peer education, peer counseling, peer monitoring, and peer assessment. For example, peer tutoring involves students alternating roles as tutors and learners, enabling their growth through guiding peers and receiving peer feedback. Peer assessment, typically conducted before final evaluations, allows students to reflect and grow by evaluating their peers and receiving feedback in return (Topping & Ehly, 2001). Thus, PAL is not a singular instructional strategy, but rather includes multiple reciprocal interactions among peers (Williams & Reddy, 2016). In this study, PAL primarily functioned as peer tutoring, with each student alternating between the roles of tutor and learner.

PAL has been acknowledged by educational researchers for its potential to motivate students to learn. From the perspective of SDT, PAL reduces students' reliance on instructors and encourages them to develop assessment and communication skills. Additionally, students generally feel connected and cared for as they mentor their peers and receive feedback from them (Nshimiyimana & Cartledge, 2020). Such PAL processes foster students' autonomy, competence, and relatedness, thereby improving their learning performances (Byl & Topping, 2023).

2.2 PAL and SDT

From the perspective of SDT, motivation is categorized into extrinsic and intrinsic motivation (Kashefian-Naeeni et al., 2024; Ryan & Deci, 2020). Extrinsic motivation refers to an individual's motivation to engage in an activity from the influence of external factors such as rewards or punishments (Kashefian-Naeeni et al., 2024). In contrast, intrinsic motivation refers to the motivation of an individual to act due to his/her interest or willingness (Hazzan-Bishara et al., 2025). Scholars have indicated that individuals are motivated to take certain actions in order to satisfy three basic

human psychological needs: autonomy, competence, and relatedness (Chiang, 2025; Feng et al., 2025); moreover, there is a linkage between these three basic human psychological needs, with autonomy as the core concept (Chiang, 2025). According to SDT, when these basic psychological needs are met, students' intrinsic and intensive learning needs can be met, and they tend to choose and monitor their own learning; thus, their competence and relatedness could be improved. That is, via meeting these basic psychological needs, students' intrinsic motivation could be enhanced, further supporting their achievement of learning goals (Chiang, 2025; Ryan & Deci, 2020).

Völlinger et al. (2023) further designed an SDT-based learning activity within PAL, aiming to enhance the effectiveness of PAL by stimulating students' intrinsic motivation. For example, students were allowed to select their reading materials to enhance their autonomy; their competence was reinforced through feedback on learning performance following reading assessments. Their relatedness was fostered through collaborative learning in small groups, with rewards for positive social interactions (Völlinger et al., 2023). The findings revealed that compared to a conventional instructional approach, SDT-integrated PAL significantly enhanced students' reading competencies and intrinsic reading motivation. Hence, leveraging SDT to design PAL activities can enhance the effectiveness of this strategy, making it a valuable direction for consideration. Furthermore, when evaluating the effectiveness of PAL, it is important to explore not only students' learning performance and intrinsic motivation but also the specific impacts on their autonomy, competence, and relatedness.

2.3 PAL in physical education

Given limited educational resources and large class sizes, PAL has received considerable attention from scholars in the context of physical education. For instance, Putri and Widiyono (2022) argued that PAL could improve the learning quality of traditional dance, enhancing students' dance skills performance. Also, Huang and Tongdechaoen (2024) applied PAL in an elementary school badminton course, and demonstrated its comprehensive effectiveness in terms of enhancing students' badminton skills through assessments of five specific badminton skills. In the same vein, PAL has been verified as positively influencing students' motor skills performance in other sports, including volleyball (Rizky et al., 2023), basketball (Mubarak et al., 2024), and tennis (Yang & Wattanapayungkul, 2024). Overall, these research findings highlight the potential of PAL in physical education.

The majority of previous studies relied on teachers providing consistent scaffolding to support students in completing peer tutoring tasks. For instance, in Putri and Widiyono's (2022) study, the teacher presented dance demonstration videos for students to observe dance skills before engaging in PAL. In Iserbyt et al.'s (2011) study, the teacher provided task cards to guide students in tennis skills training. Similarly, Lentillon-Kaestner and Roure (2024) prepared observation checklists to assist peer tutors in giving feedback to learners based on specific criteria. These studies indicated that students did not receive adaptive scaffolding during the PAL process. Moreover, there is limited discussion in the literature on how to guide or support peer tutors in providing adaptive instruction and feedback during PAL. Students are also

concerned about the capability of inexperienced peer tutors to effectively guide others and assist them in addressing individual learning challenges (Nshimiyimana & Cartledge, 2020; Swallow et al., 2023). For example, a study investigating the quality of peer assessment provided by students revealed that students' subjectivity was the main factor contributing to the discrepancy in assessment results (Roberts et al., 2024). Therefore, the central focus of this study was how to guide students to provide adaptive feedback from a professional and objective perspective to help peer learners improve their learning performance.

In addition, most of the previous PAL studies specified that students can benefit from group activities (Coliñir et al., 2022), but the methods of grouping were not consistent. The decision of whether to group students based on their abilities or friendships is also crucial. Scholars have suggested that, in physical education, the implementation of PAL should enable students to choose their own group members. Allowing students to select their own group members not only helps build positive peer relationships, but also promotes their competence and relatedness, thereby positively influencing their learning motivation. This method is also more time-efficient compared to other grouping strategies (Lentillon-Kaestner & Roure, 2024).

Therefore, for the present study, we developed an AI-facilitated PAL system to provide students with adaptive feedback similar to that from a professional teacher, thereby supporting their interactions during the PAL process. Through the provision of assessment criteria and corresponding descriptive examples, students were guided to learn to conduct PAL from an objective perspective. Based on successful experiences from past studies, students were allowed to select their own PAL group members to create a more motivated learning environment. To determine whether the previously identified learning benefits associated with PAL became more pronounced with the AI-facilitated PAL system, this study explored students' motor skills performance, competence, relatedness, and learning motivation.

2.4 AI-facilitated PAL

An intelligent tutoring system (ITS) equipped with AI can support automated assessment and provide adaptive feedback, creating innovative environments to enhance students' learning experiences (Deeva et al., 2021; Kohnke, 2023; Mousavinasab et al., 2021; Ogata et al., 2024; Sembey et al., 2024). Hence, some scholars have advocated for the use of an ITS to support students in providing peer feedback with higher quality, thereby optimizing their PAL experiences (Darvishi et al., 2022; Lin et al., 2024).

There are numerous successful examples of such applications across different academic disciplines. For instance, Walker et al. (2014) developed an ITS that provided adaptive prompts and feedback during peer tutoring, which effectively enhanced high school students' mathematics learning performance. Haq et al. (2020) developed an intelligent tutoring-supported collaborative learning (ITSCL) system to assist students in co-constructing knowledge during PAL activities. Initially, students were required to complete individual learning tasks and present their insights for subsequent learning activities. Next, they provided feedback and ratings on their peers' knowledge and insights, facilitating their self-reflection. Finally, the system analyzed

each student's knowledge contributions, identified the best responses, and presented the results to all students. This study showed that the system effectively supported PAL activities among peers, contributing positively to their academic performance. Scholars have further pointed out that students can collectively change their approach to problem solving with the support of AI (Güner & Er, 2025); for example, they can use AI in a complementary way to improve their own competence by integrating AI's feedback into their existing knowledge (Güner & Er, 2025), or they can reflect on the feedback provided by AI in order to improve the quality and quantity of their peer feedback (Guo et al., 2024; Sembey et al., 2024). These findings illustrate how the effective integration of AI into educational settings can further increase the benefits of PAL.

Additionally, Han et al. (2021) developed a learning analytics dashboard. According to students' learning status, it provided adequate and immediate feedback for students during face-to-face argumentative activities. The study indicated that visual feedback and adaptive support based on learning analytics positively influenced students' learning interest and outcomes. Lee and Kim (2024) further illustrated that the feedback provided on learning analytics dashboards could help learners to make decisions on their own, and hence enhance their autonomy, the objective of SDT. This perspective supports the use of AI to provide automated assessment and the visualization of learners' status in the learning analytics dashboard. This implies that the AI-facilitated PAL approach has the potential to satisfy the autonomy objective of SDT, and to enable students to achieve better learning outcomes.

Nonetheless, the aforementioned research primarily focused on subject-based curricula. A literature review on ITS revealed that most studies have concentrated on computer science education, medical theoretical education, and mathematics (Alra-khawi et al., 2023; Mousavinasab et al., 2021). The role of ITS in facilitating PAL within skill-based courses remains underexplored. Furthermore, most existing studies paid attention to supporting students' PAL within online learning environments (e.g., Haq et al., 2020; Walker et al., 2014). PAL could be potentially beneficial for skill-based, face-to-face learning environments, particularly in physical education. In-person interaction enables students to engage in peer tutoring through both verbal and non-verbal means, such as using physical contact to adjust peers' poses, and providing visual feedback through demonstrations. Therefore, the ITS developed in this study focused on enhancing the quality of peer tutoring and feedback in face-to-face learning environments.

Lastly, most ITS applications aimed at enhancing PAL predominantly utilize natural language processing (NLP) techniques, analyzing students' written messages (e.g., Haq et al., 2020; Walker et al., 2014). In this study, the system was applied in physical education, where text-based communication alone was neither intuitive nor sufficient for learning motor skills. Human activity analysis was adopted as the primary technique for automated assessment.

2.5 Human activity analysis in physical education

Human activity analysis can assist participants in completing rehabilitation tasks, can monitor hazardous human activities in public spaces, and can aid in the development

of autonomous driving systems, making it one of the more challenging and significant fields within AI (Sardari et al., 2023). Of particular relevance to this study is the use of skeleton data analysis to support physical training tasks, which has demonstrated several advantages (Sardari et al., 2023). For instance, employing OpenPose to collect skeleton data and applying model training to construct an ITS have been recognized by many physical education professionals as an effective approach (e.g., Bergner et al., 2021; Kang et al., 2023; Lin et al., 2023; Pan, 2024).

OpenPose is an open-source real-time system capable of recognizing human key points, and has been widely applied in various tasks for human activity analysis (Cao et al., 2019). It effectively addresses the issue of overlapping skeletal joints in pose recognition (Kim et al., 2021; Lin et al., 2023). Moreover, compared to conventional algorithms, OpenPose demonstrates superior extraction accuracy and requires less computational time (Jin et al., 2022). As pointed out by Jin et al. (2022), a dance teaching system using OpenPose for recognition achieved a maximum accuracy rate of 96% across eight different dance movements. This high level of accuracy illustrated the effectiveness of OpenPose in terms of recognizing motor skills.

Leveraging the advantages of OpenPose, many researchers have used it for pose recognition and adaptive training intensity or feedback, contributing to the development of an ITS that effectively enhances students' growth in motor skills. For example, Pan (2024) integrated OpenPose with virtual reality (VR) to develop a VR training system for sports games. It evaluated students' skill levels and provided personalized training scenarios of varying degrees of difficulty to ensure that students with different prior abilities could benefit from the training. The results showed that the system demonstrated outstanding performance in accuracy, recall, and F1-score in university teaching settings. Also, this teaching method helped students gain a comprehensive understanding of sports competition training, leading to improved training outcomes. In addition, grounded in embodied cognition theory, Bergner et al. (2021) utilized OpenPose to collect dance movement data from students and provide analytic feedback for students. The findings showed that the system enabled students to use their own dance experiences to grasp abstract concepts in data science, thereby enhancing their learning motivation. Lin et al. (2023) employed OpenPose to develop an ITS for a university badminton course. By providing students with immediate automated movement assessments along with practice video reviews, the system successfully facilitated students' skills learning performance when compared to conventional teaching methods.

The aforementioned research findings not only demonstrate the popularity of OpenPose in motor skill recognition, but also highlight the applications and value of human activity analysis in physical education. As a result, this study drew upon these successful experiences to develop an ITS utilizing OpenPose as the recognition technology. The system incorporated features such as automated pose assessment and practice video reviews. Additionally, common metrics for ITS, including accuracy, recall, and F1-score, were examined to evaluate the system performance. Several studies have also investigated the effects of an ITS on students' motor skills performance and learning motivation when applied to physical education (e.g., Bergner et al., 2021; Lin et al., 2023; Pan, 2024). To further investigate the potential of an ITS, in

addition to learning performance and motivation, this study explored the effects of an ITS combined with SDT theory on students' autonomy, competence, and relatedness.

2.6 Challenges of human activity analysis in physical education

Previous scholars have identified several unresolved issues when applying human activity analysis in physical education and rehabilitation training. For instance, some users struggle to fully understand how to adjust their poses based on the automated assessments provided by the system. They require additional explanations, suggestions, or examples to establish the direction and strategies for correction (Lin et al., 2023; Sardari et al., 2023). To address this issue, some scholars have advocated for enhancing the collaboration between teachers and AI. In the study by Kang et al. (2023), students received two types of feedback. The first type was automated feedback from AI, which included synchronized dance video reviews between students and the teacher, similarity scores, and prompts for body movements that required modification. The second type of feedback was provided by the teacher based on the content of the videos submitted by the students. The results revealed that the feedback from both the teacher and AI was complementary, providing guidance from different perspectives. In addition, Zhou (2024) utilized OpenPose to assess students' poses during sports training, providing valuable information for the teacher to enhance instructional practices. The results indicated that applying human activity analysis in physical education could assist the teacher in strengthening teaching efforts, ultimately improving students' mastery of basketball skills. However, the study primarily focused on the introduction of the algorithm, providing limited details on the implementation processes and methods in the teaching context. The specific practical applications of the system in educational settings remained unclear.

Above all, when implementing an ITS in physical education, attention must be paid to the fact that some students may struggle to understand and effectively utilize the feedback provided by the system. Having teachers enhance the feedback or identify AI-generated feedback before providing guidance could be viable solutions. Yet, it is crucial to note that students still want immediate insights into their learning performance prior to teacher intervention (Lin et al., 2023). Thus, drawing on the recommendations of Sardari et al. (2023) and Lin et al. (2023), the present study adopted visual representations to highlight the poses that required improvement, provided individual scores for each pose, and included the explanation of these scores to improve the readability of the AI feedback. Furthermore, previous research generally focused on applying AI to provide adaptive and instant feedback to assist individual students during motor skill practicing processes (e.g., Lin et al., 2023; Sardari et al., 2023), while neglecting the provision of support for group learning activities, especially using AI to facilitate PAL in motor skill training. To address this issue, the present study employed PAL to integrate peer feedback into the learning activity. This aimed to address the challenges some students face in understanding AI feedback without increasing the teacher's workload (Lin et al., 2023). Through this learning strategy, students received feedback from three different sources: AI, the teacher, and the peer, which has significant potential for enhancing students' motor skill acquisition (Kang et al., 2023). Such integration also echoes the social constructivist

proposition that students co-construct the knowledge needed to learn a skill through interactions with peers and ITS. For example, students will express their suggestions, receive feedback from peers, and communicate and negotiate their perspectives during PAL (Williams & Reddy, 2016), while the ITS serves as complementary scaffolding, providing immediate personalized feedback on different facets of the process, and visual dashboards to illustrate individual performance. This scaffolding can be used to clarify misunderstandings or support correct perspectives, complementing peer-to-peer dialogues and enhancing the quality of PAL (Guo et al., 2024; Sembey et al., 2024). Therefore, AI-supported PAL is not only theoretically sound, but is also a strategy for further active implementation in the teaching field.

3 The AI-PAL mode

Previous studies have pinpointed that integrating basic psychological needs from SDT (i.e., autonomy, competence, and relatedness) as motivational elements into PAL can stimulate students' intrinsic motivation, leading to improved learning outcomes (Völlinger et al., 2023). In particular, allowing students the freedom to choose their own group members based on the level of friendship would more positively fulfill students' basic psychological needs (Lentillon-Kaestner & Roure, 2024). Furthermore, an increasing number of scholars have indicated that properly utilizing AI could promote the quality of feedback provided by students to peers, resulting in better learning outcomes (Darvishi et al., 2022; Guo et al., 2024; Sembey et al., 2024). Hence, drawing on the successful experiences of previous scholars, the present study incorporated SDT to propose the AI-PAL model (see Fig. 1), and allowed students to choose their group members by referring to the literature. Also, an AI-facilitated PAL system was developed to assess students' motor skill performance and provide adaptive feedback as scaffolding to support students in executing peer tutoring tasks within a collaborative learning environment.

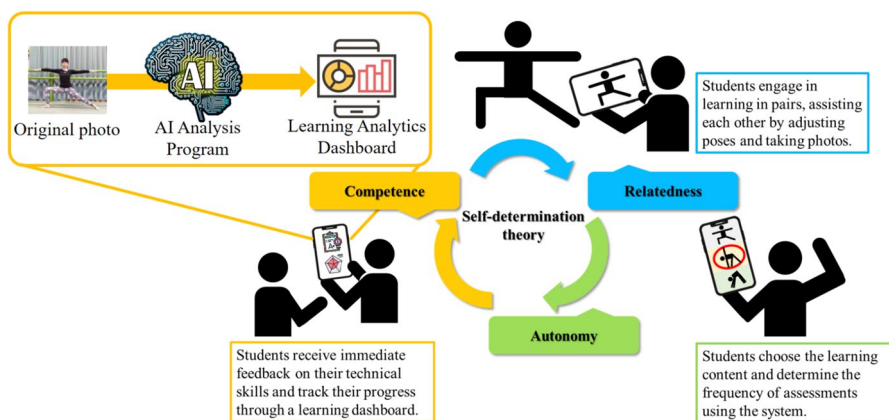


Fig. 1 The AI-PAL mode

As shown in Fig. 1, regarding relatedness, students engaged in group learning by adjusting each other's poses and taking photos. In terms of autonomy, students chose the order in which they learned different poses and determined the number of assessments using the ITS. In terms of competence, they received feedback on their motor skill assessments from the system. This provided them with a clear understanding of whether their learning performance was continuously improving, being maintained, or deteriorating through a dashboard.

In the AI-facilitated PAL system, the responsive web design (RWD) was employed in the front-end interface to ensure that students could use it on various devices and have a better user experience. The primary function of the system was to provide immediate analysis of students' motor skill performance, providing feedback through visual representations. The overall system structure is illustrated in Fig. 2. The following sections detail the data pre-processing, knowledge model construction, knowledge model validation, and system functionalities.

3.1 Data pre-processing

The training photos of somatic practice poses embedded in the AI-facilitated PAL system were class assignments completed by university students over the years. The participating students, aged between 18 and 20, were informed of the precautions before taking the photos. They were instructed to ensure that the poses were fully captured in the photos. This would minimize errors arising from improper shooting, which could hinder the accurate analysis and evaluation of pose quality. After collection, incomplete or unanalyzable photos were discarded, resulting in a total of 1,795 photos used for knowledge model construction.

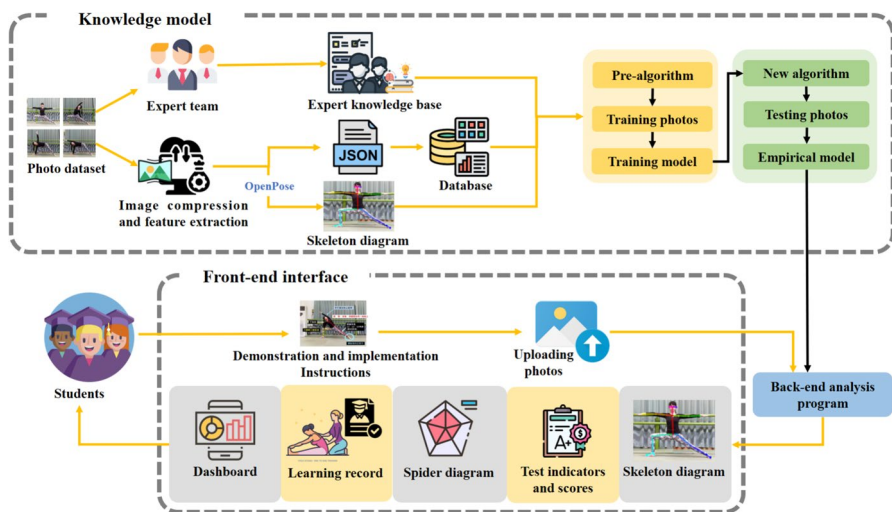


Fig. 2 Structure of the AI-facilitated PAL system

3.2 Knowledge model construction

The photo dataset was evaluated by both an expert team and a system. The expert team consisted of three experienced somatic practice teachers with valid somatic practice certification. A group meeting was held to establish four to six test indicators for each somatic practice pose, with each indicator rated on a scale from 1 (minimum) to 3 (maximum). Then, each member of the expert team rated every photo based on the established test indicators individually. The Kendall's ω value was 0.724, indicating a high level of consistency among the raters. Afterwards, the expert knowledge extraction technique was employed to integrate the ratings from the expert team, resulting in the establishment of an expert knowledge base. Finally, based on the expert knowledge base, formulas for calculating key point angles or slopes were developed to complete the construction of the pre-algorithm (see Fig. 3).

In terms of the system's evaluation, following the recommendations of Lin et al. (2023) and Pan (2024), OpenPose, developed by Carnegie Mellon University, was adopted for pose recognition. First, the photo dataset underwent image compression and feature extraction. This was necessary because most of the photos were taken at a resolution exceeding 1920×1080 . Directly uploading these photos would result in excessive hardware load, slowing down the analysis speed and indirectly increasing the likelihood of analysis errors and unanalyzable images. Therefore, this study followed the recommendations of Nakano et al. (2020) by compressing the photo resolution to 192×128 to enhance the accuracy of labeling the skeleton diagrams. In addition, feature extraction was conducted to identify the primary areas of analysis within each photo, which was then imported into OpenPose for processing and label-

Algorithm yoga rule;

Given : Json file of training photos

Result : yoga rule

Begin

1. If countable items unable to calculate > 3 then return score
 $= [-1, -1, -1, -1, -1, -1, -1]$
2. For enter this analytic indicators score into an array
 $A = \{\text{analytic indicators score}\}$
3. For count analytic indicators keypoint
 $B = \{\text{analytic indicators keypoint}\}$
4. If analytic indicators keypoint(A) = 2 then Cal(angle)
5. Else if analytic indicators keypoint(A) = 3 then Cal(slope)
6. C=1;
7. If angle or slope calculated then
 For Enter into array
 $F = \{\text{analytic indicators angle or slope}\}$
 C++;
8. D++;
9. Repeat step 1 to 7 until $D > \text{count(pic)}$

End

$$\text{Slope} = \frac{Y_1 - Y_2}{X_1 - X_2}$$

$$\text{Angle} = \frac{\vec{AB} \cdot \vec{AC}}{|\vec{AB}| |\vec{AC}|}$$

Fig. 3 Pre-algorithm and formulas

ing. After processing the photos with OpenPose, two types of data were generated: a skeleton diagram that recorded the key points, and a JSON file that documented the coordinates of the key points. The JSON files were compiled into a dataset to be input into the pre-algorithm. Along with the aforementioned 1,795 annotated skeleton diagrams, the pre-algorithm was employed to train the model.

The deep learning model used in this study was based on a convolutional neural network (CNN) and was programmed using Python's Keras Sequential API. The primary CNN architecture comprised three layers: convolutional layers, fully connected layers, and pooling layers (Fukushima, 1980). The model employed Rectified Linear Units (ReLU) to activate $16\ 3 \times 3$ filters, which were connected to a max pooling layer. This design primarily served to reduce unnecessary layers, mitigate gradient vanishing, and enhance computational efficiency. A convolutional layer was then applied to extract key point features and calculate the relative distances, angles, and slopes between each key point. Additionally, a fraction of the weights was randomly dropped to prevent overfitting. The pooling layers were subsequently connected to the fully connected layers, and batch normalization was applied to accelerate convergence and enhance computational performance. Finally, the model was compiled using Keras and Theano, employing categorical cross-entropy as the loss function (Sun et al., 2016). The loss function was primarily used to measure the difference or error between the model's predictions and the actual target, aiding in the adjustment of weights and parameters. Minimizing the value of the loss function could improve the model's accuracy and precision. For parameter settings, the initial learning rate was set to 0.0001, with an epoch value of 20 and a batch size of 20. The Adam optimizer without decay was adopted to control the learning rate, with a β value of 0.9. The hardware used during the development process included an Intel Core i7-9700 K processor, 32 GB RAM, an Nvidia GTX-2060TI GPU, and the Windows 10 operating system. After completing model training, sections requiring adjustment within the pre-algorithm were recompiled and processed, forming a new algorithm.

Table 1 presents the test indicators for each pose within the expert knowledge base, along with the scoring ranges calculated using the new algorithm. Taking the Warrior II pose as an example (see Fig. 4), one of the test indicators was that the arms are extended straight and parallel to the shoulders. The coordinates of key points A, B, and C were then captured and calculated using the angle formula, resulting in an angle of 179.59 degrees. Referring to the expert knowledge base, the scoring range for this test indicator indicated a score of 3.

3.3 Knowledge model validation

To examine the new algorithm, an additional set of 1,372 practice photos was collected from university students aged 18 to 20 enrolled in a physical education course, in addition to the previously mentioned photo dataset. Of these, 678 photos were analyzed by the system and the expert team. Kendall's ω values ranged from 0.708 to 0.818, indicating a high level of consistency between the system and the expert team in their ratings. This validated that the model developed in this study could successfully identify the quality of students' poses and provide scores and feedback comparable to those of the experts.

Table 1 Test indicators of somatic practices

Pose	Test indicator	Score		
		1	2	3
Warrior II	The head and pelvis are aligned on the same vertical line. ^a	< 154	155~164	165~180
	The arms are extended straight and parallel to the shoulders. ^a	< 154	155~164	165~180
	The thighs are positioned perpendicular to the calves. ^a	< 74	75~84	85~95
	The knees are fully extended. ^a	< 154	155~164	165~180
	The toes point straight ahead. ^b	> 0.75	0.51~0.74	< 0.5
Extended side angle	The head, pelvis, and heels are aligned on the same vertical line. ^a	< 154	155~164	165~180
	The arms are extended straight and angled diagonally upward. ^a	< 154	155~164	165~180
	The shoulders are not shrugged. ^a	< 154	155~164	165~180
	The thighs are positioned perpendicular to the calves. ^a	< 74	75~84	85~95
	The knees are fully extended. ^a	< 154	155~164	165~180
Revolved side angle	The toes point straight ahead. ^b	> 0.75	0.51~0.74	< 0.5
	The arms are extended straight and angled diagonally upward. ^a	< 154	155~164	165~180
	The shoulders are not shrugged. ^a	< 154	155~164	165~180
	The thighs are positioned perpendicular to the calves. ^a	< 74	75~84	85~95
	The knees are fully extended. ^a	< 154	155~164	165~180
Triangle	The hands rest on the thighs. ^a	< 133	134~140	> 141
	The toes point straight ahead. ^b	> 0.75	0.51~0.74	< 0.5
	The hands are perpendicular to the ground, with palms facing forward. ^a	< 154	155~164	165~180
	The head and pelvis are aligned on the same vertical line. ^a	< 154	155~164	165~180
	Both knees are fully extended. ^a	< 154	155~164	165~180
Revolved triangle	The toes point straight ahead. ^b	> 0.75	0.51~0.74	< 0.5
	The arms are extended straight and close to the ears. ^a	< 160	161~167	> 168
	The head and pelvis are aligned on the same vertical line. ^a	< 154	155~164	165~180
	Both knees are fully extended. ^a	< 154	155~164	165~180
	The toes point straight ahead. ^b	> 0.75	0.51~0.74	< 0.5

angle: a; slope: b

Subsequently, the empirical model was integrated into the backend analysis program, utilizing the remaining 694 photos for validation of the model's effectiveness and reliability. This study followed the recommendations of Ashwin et al. (2023) by employing a confusion matrix to compute four commonly used metrics for evaluating system performance: accuracy, precision, recall, and F1-score. The confusion matrix was composed of a 2×2 matrix, including the four elements of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Specifically, TP indicated that the system provided correct feedback for accurately identified photos, while TN showed that the system provided an incorrect notification for misidentified photos. FP denoted the system giving correct feedback for misidentified photos,

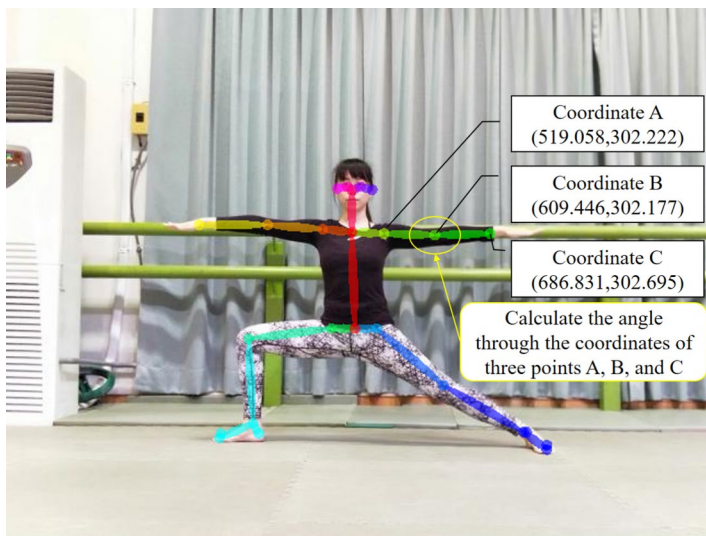


Fig. 4 One of the test indicators for the Warrior II pose

$$\begin{aligned}
 \text{Precision} &= \frac{TP}{TP + FP} \times 100\% & \text{Accuracy} &= \frac{TN + TP}{TP + FP + FN + TN} \times 100\% \\
 \text{Recall} &= \frac{TP}{TP + FN} \times 100\% & \text{F-1 score} &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}
 \end{aligned}$$

Fig. 5 Formulas for the four metrics

while FN represented the system providing an incorrect notification for accurately identified photos. The formulas for the four metrics are illustrated in Fig. 5.

After each photo was uploaded, the model processed it and output a score and feedback to the user interface in approximately 2.954 s. The overall accuracy exceeded 98%, with precision, recall, and F1-score all surpassing 97%. This validated that the model developed in this study was both effective and reliable.

3.4 System functionalities

In the photo upload interface, students first viewed a demonstration of the poses by the teacher, as well as instructions to enhance their understanding of how to perform the poses correctly. Afterwards, they uploaded practice photos of their poses for intelligent assessment, allowing them to receive immediate feedback from the system. For example, the system displayed a skeleton diagram of the human body for the practiced pose, along with a description of the score level, total scores, test indicators and scores, and a spider diagram of scores. One type of personalized feedback of interest is the Skeleton diagram, which is an overlay of human joints and bones using

different colored lines and dots in the exercise photos uploaded by the students. The diagrams are used to assist students in recognizing whether or not the angle of execution of a pose is correct based on the feedback of the colored lines. Test indicators and scores were pre-set by professional teachers for each pose to provide students with a professional-oriented assessment; at the same time, AI calculated test scores for each indicator based on the student's performance. The AI also checked the scores of each assessment indicator according to the student's performance, so that they could make targeted improvements according to the items with lower scores to improve their learning efficiency. The Spidergram of scores is a visualization of test indicators and scores. By printing the scores of all the test indicators on a zero-centered scale, students can quickly and intuitively understand their strengths and weaknesses in each test indicator. Students could refer to this feedback from the system to engage in PAL (see Fig. 6). For example, based on the feedback provided by the system, each student can independently decide whether he/she wants to practice the same pose again, switch to a different pose, or review the relevant teaching examples and instructions again. At the same time, the peer tutor can also refer to this personalized feedback to provide more specific suggestions and ideas to help the student focus on his/her own weaknesses and set up a follow-up practice plan.

In the learning record interface, students could review the timestamps of their previously uploaded practice photos along with the corresponding total scores. They could also access comprehensive feedback for each specific practice pose, allowing them to thoroughly compare and analyze the relationship between their past poses and the feedback provided by the system. Students could observe their learning progress over time, from past to present. It encourages students to choose whether they want to provide peers with feedback or seek feedback from peers to improve deficiencies based on their needs during the learning progress. This means that students are able to switch autonomously between the tutor and learner roles, taking into account the analyzed information provided by the system.



Fig. 6 PAL and the feedback interface

In the dashboard interface, in addition to displaying the best score for each pose, the scoring data were presented visually. Specifically, for each pose, the student's score was converted into percentages and was shown as a bar chart to illustrate the gap between their current score and the maximum possible score. This allowed students to understand their learning progress for each pose from various perspectives, including scores, percentages, and bar charts (see Fig. 7). At the same time, it highlights the postures that are consistently lower in correctness. This encourages students to prioritize these postures in subsequent practice, and encourages peer tutors to pay attention to the postures and to provide more specific assistance to help peer learners overcome their learning difficulties.

4 Research design

4.1 Participants

In this study, convenience sampling was used to recruit participants for the experiment based on the willingness of students in the selected classes. A total of 90 s-year students enrolled in a physical education course at a technological university in central Taiwan were recruited for this study. Their ages ranged from 18 to 20; 55.56% were females and 44.44% were males. Most of the students were beginners in somatic practices and had not received professional somatic training before. One class of 44 students was designated as the experimental group, adopting the AI-PAL mode. The other class of 46 students served as the control group, adopting the C-PAL mode. The teaching progress and content for both classes were identical and were delivered by the same experienced teacher. For the in-class learning activity, students selected their partners and worked in pairs to collaboratively engage in PAL tasks.

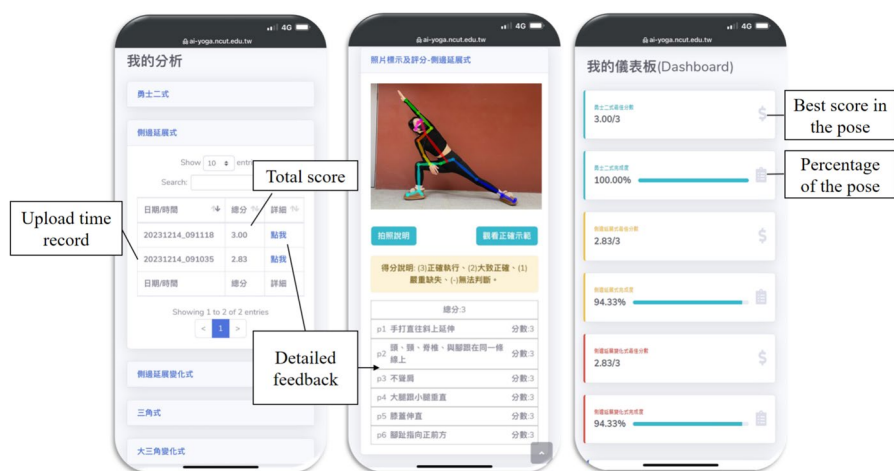


Fig. 7 Interfaces of the learning record and dashboard

4.2 Experimental procedure

The experimental procedure is shown in Fig. 8. The experiment lasted a total of 11 weeks and included two instructional phases: Phase A (4 weeks for 400 min) and Phase B (4 weeks for 400 min). The learning content was somatic practices, which encompass a variety of forms of exercise that emphasize guiding participants to reflect on their own exercise habits and to pay attention to the sensations of the body and mind during exercise. The practices include yoga to enhance participants' awareness of their bodies for the purpose of improving their quality of life (Galantino et al., 2019; Meehan and Carter, 2021). While yoga is a commonly recognized form within this domain, the term "somatic practice" is used in this study to foreground the educational focus on reflection and self-regulation, and to avoid cultural or spiritual connotations that may not align with the instructional objectives, as suggested by Green (2002). This terminology aligns with this study's pedagogical goals rooted in self-regulated learning.

During Phase A, both groups adopted a conventional teaching mode to learn the identical set of poses. The teacher first provided a verbal explanation and demonstration of the poses and then led the students in practical exercises. During the practice sessions, the teacher observed the students' performance and selected a few to demonstrate their poses in sequence, providing individualized analysis, guidance, and adjustments. This was intended to help students gradually develop their abilities to observe, analyze, and assist others in improving the quality of their poses.

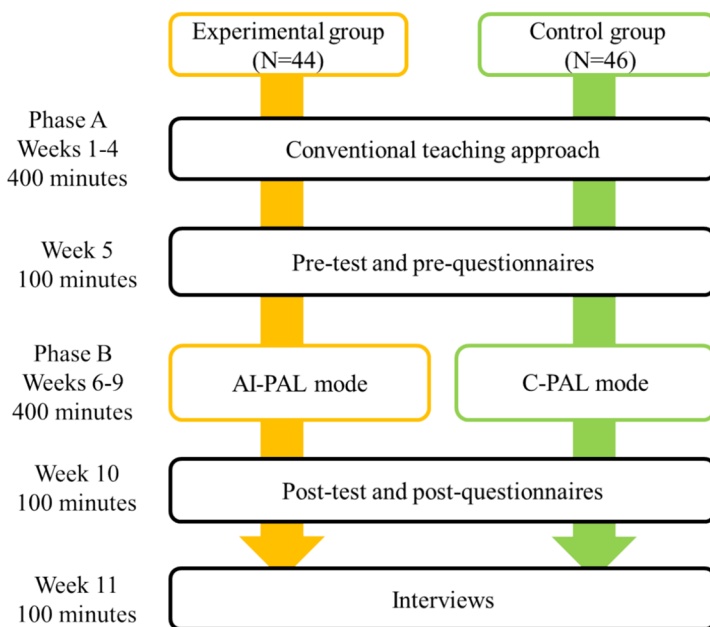


Fig. 8 Experimental procedure

Upon completing Phase A, all students took a pretest of motor skills and completed pre-questionnaires on learning motivation and basic psychological needs. Subsequently, the two groups adopted different modes of learning.

In Phase B, the learning content and process for both groups remained consistent. The teacher began with a lecture and demonstration of the poses, guiding the students through practical exercises to help them understand the fundamental concepts and skills for performing the poses. Students then engaged in PAL in pairs, taking turns assuming the roles of tutor and learner. This collaborative learning approach aimed to enhance the quality of their poses. The teacher moved around the classroom, providing guidance and feedback to each group. The theoretical difference between the two groups is that the AI-PAL mode is based on the theoretical foundation of SDT, and applies AI to provide personalized feedback as a scaffold to more actively meet students' basic psychological needs, whereas the C-PAL mode practices the traditional PAL mode and lacks the support of personalized feedback provided by AI. To better illustrate the instructional structure of each condition, Table 2 outlines the theoretical and practical distinctions between the AI-PAL and C-PAL modes to clarify their differences.

Regarding the PAL activity, the experimental group adopted the AI-PAL mode for learning. During the process of guiding each other, students could take photos of their partner's poses and upload them to the learning system. After receiving the assessment results provided by the system, students could refer to the feedback for further discussion and practice or choose to conclude their session.

On the other hand, the control group adopted the C-PAL mode. During the process of guiding each other, students could use their mobile phones to take photos of their partner and discuss the poses based on the test indicators, which were identical to those in the ITS. Finally, each student was required to upload the practice photos to the online learning system, where the teacher would provide feedback within a week.

Upon completing Phase B, all students took a posttest of motor skills and completed post-questionnaires on learning motivation and basic psychological needs. In the following week, all students participated in semi-structured focus group interviews in groups of five to six, each lasting approximately 10 min.

4.3 Instruments

As recommended by the teacher, the scoring rubrics for motor skills performance were based on Hsia and Hwang (2020). Technical skills were selected as the evalu-

Table 2 Comparison of the AI-PAL and C-PAL models in theory and practice

	AI-PAL	C-PAL
Theoretical basis	SDT	Traditional PAL
Source of feedback	AI feedback and peer feedback	Teacher feedback and peer feedback
Provided support	learning record (e.g., uploaded practice photos) and personalized feedback (e.g. skeleton diagram, score level, spider diagram) from AI	learning record (e.g., uploaded practice photos) and feedback from the teacher

ation dimensions in alignment with the teaching content. The scoring scale ranged from a minimum of 1 to a maximum of 4. Two somatic practice teachers with more than 7 years of teaching experience were invited to conduct the evaluation. The Spearman's correlation coefficient for the scoring results was 0.77 ($p < 0.01$), indicating a high level of consistency between the raters. The average score given by the two teachers was used as the student's motor skills performance score.

The learning motivation questionnaire was adapted from Pintrich et al. (1991) and was slightly modified to fit the learning context of this study. It consisted of six items, including three for intrinsic motivation and three for extrinsic motivation. The questionnaire adopted a 5-point Likert scale (1 = *strongly disagree*; 5 = *strongly agree*). For instance, one of the items for intrinsic motivation was "In the physical education course, I prefer learning poses that arouse my curiosity, even if they are difficult to learn." One of the items for extrinsic motivation was "If possible, I want to get better grades in this physical education course than most of the other students." The Cronbach's alpha value was 0.88.

The basic psychological needs questionnaire was adapted from Vlachopoulos and Michailidou (2006) and consisted of 11 items. It included four items for autonomy (e.g., "I feel very strongly that I have the opportunity to choose my preferred way of somatic practices"), four items for competence (e.g., "I feel that I can manage the requirements of the somatic practices"), and three items for relatedness (e.g., "I feel there are open channels of communication with my group members"). This questionnaire adopted a 5-point Likert scale (1 = *strongly disagree*; 5 = *strongly agree*). The Cronbach's alpha values were 0.91, 0.92, and 0.95, respectively.

The semi-structured interview questions were adapted from Hwang et al. (2009) and were slightly modified to fit the learning contexts of the experimental group and the control group. There were seven questions in total, for instance "What part of learning somatic practices in this way do you think helped you get the most out of it? What part did you learn the most from? Please give specific examples."

4.4 Data analysis

Several scholars have pointed out the importance of ruling out differences in students' starting points in assessing the impact on students' performance and perceptions of learning after the intervention of the instructional strategies (Durak, 2025; Lin et al., 2020, 2023). To precisely explore whether different learning modes impact students' motor skills performance, learning motivation, autonomy, competence, and relatedness between the two groups, the present study employed an analysis of covariance (ANCOVA) with a significance level set at $p < 0.05$. The data were analyzed using SPSS. When significant differences were found between the groups, effect sizes were subsequently calculated.

The interview process was audio recorded and transcribed verbatim by the researchers. The analysis procedure followed the steps outlined by Creswell (2013). Based on the most frequently mentioned content by participants, the first and second authors initially identified two themes: enhancing learning effectiveness and improving the quality of PAL. Afterwards, the two researchers reviewed the transcripts independently, further refining similar content and categorizing it under consistent codes.

Table 3 ANCOVA results of learning performance

Group	<i>N</i>	Mean	<i>SD</i>	Adjusted Mean	<i>SE</i>	<i>F</i>	η^2
Experimental group	44	2.33	0.38	2.33	0.05	7.81**	0.08
Control group	46	2.12	0.33	2.13	0.05		

** $p < 0.01$ **Table 4** ANCOVA results of learning performance

Group	<i>N</i>	Mean	<i>SD</i>	Adjusted Mean	<i>SE</i>	<i>F</i>	η^2
Intrinsic motivation							
Experimental group	44	3.82	0.84	4.03	0.12	4.88*	0.05
Control group	46	3.83	0.88	3.63	0.12		
Extrinsic motivation							
Experimental group	44	3.88	0.77	3.97	0.13	3.16	
Control group	46	3.72	0.92	3.64	0.13		
Overall							
Experimental group	44	3.85	0.75	4.01	0.12	4.94*	0.05
Control group	46	3.78	0.87	3.62	0.12		

* $p < 0.05$

The Cohen's kappa value was 0.83, indicating a high level of consistency between the two coders.

5 Results

5.1 Analysis of learning performance

After performing a homogeneity test for regression coefficients, the F -value did not reach a significant level ($F=0.30$, $p=0.59 > 0.05$), suggesting that the learning performance of the two groups did not violate the assumption of homogeneity. Thus, ANCOVA could be employed. As shown in Table 3, the experimental group had significantly higher learning performance than the control group ($F=7.81$, $p=0.01 < 0.05$), with a moderate effect size ($\eta^2=0.08$; Cohen, 1988). The adjusted mean for the experimental group was 2.33, while that for the control group was 2.13, indicating that students adopting the AI-PAL mode outperformed those adopting the C-PAL mode in terms of motor skills performance.

5.2 Analysis of learning motivation

After performing a homogeneity test for regression coefficients, the F -values for intrinsic motivation, extrinsic motivation, and overall learning motivation did not reach a significant level ($F=0.51$, $p=0.48 > 0.05$; $F=2.66$, $p=0.11 > 0.05$; $F=1.46$, $p=0.23 > 0.05$). This implied that the learning motivation of the two groups did not violate the assumption of homogeneity. Hence, ANCOVA could be employed. As shown in Table 4, the experimental group had significantly higher intrinsic motivation and overall learning motivation than the control group ($F=4.88$, $p=0.03 < 0.05$;

$F=4.94$, $p=0.03<0.05$), with small effect sizes ($\eta^2=0.05$; $\eta^2=0.05$; Cohen, 1988). Regarding intrinsic motivation, the adjusted mean for the experimental group was 4.03, while that for the control group was 3.63. As for overall learning motivation, the adjusted mean for the experimental group was 4.01, while that for the control group was 3.62. There was no significant difference in extrinsic motivation between the two groups ($F=3.16$, $p=0.08>0.05$). This indicated that the AI-PAL mode significantly enhanced students' overall learning motivation, particularly in terms of intrinsic motivation.

5.3 Analysis of basic psychological needs

After performing a homogeneity test for regression coefficients, the F -values for autonomy, competence, and relatedness did not reach a significant level ($F=3.98$, $p=0.05>0.05$; $F=0.98$, $p=0.33>0.05$; $F=0.98$, $p=0.33>0.05$). This suggested that the basic psychological needs of the two groups did not violate the assumption of homogeneity. Therefore, ANCOVA could be employed. As shown in Table 5, the experimental group had significantly higher levels of autonomy, competence, and relatedness than the control group ($F=5.23$, $p=0.03<0.05$; $F=7.02$, $p=0.01<0.05$; $F=7.02$, $p=0.01<0.05$), with medium effect sizes ($\eta^2=0.06$; $\eta^2=0.08$; $\eta^2=0.08$; Cohen, 1988). Regarding autonomy, the adjusted mean for the experimental group was 3.92, while that for the control group was 3.51. As for competence, the adjusted mean for the experimental group was 4.15, while that for the control group was 3.65. In terms of relatedness, the adjusted mean for the experimental group was 4.15, while that for the control group was 3.65. The results showed that the AI-PAL mode significantly enhanced students' autonomy, competence, and relatedness when compared to the C-PAL mode.

5.4 Interview results

The interview results revealed two advantages most frequently mentioned by students: enhancing learning effectiveness and improving the quality of PAL (see Table 6).

Table 5 ANCOVA results of basic psychological needs

Group	<i>N</i>	Mean	<i>SD</i>	Adjusted Mean	<i>SE</i>	<i>F</i>	η^2
Autonomy							
Experimental group	44	3.82	0.79	3.92	0.13	5.23*	0.06
Control group	46	3.60	0.90	3.51	0.12		
Competence							
Experimental group	44	4.03	0.78	4.15	0.13	7.02*	0.08
Control group	46	3.76	0.97	3.65	0.13		
Relatedness							
Experimental group	44	4.03	0.78	4.15	0.13	7.02*	0.08
Control group	46	3.76	0.97	3.65	0.13		

* $p<0.05$

Table 6 Themes, codes, and numbers of occurrences of the interview results

Theme	Code	The number of times mentioned	
		Experimental group	Control group
Enhancing learning effectiveness	Mastering technical skills through system feedback/ technology assistance	42	12
	Mastering technical skills through teacher feedback	1	31
Improving the quality of PAL	Enhancing the quality of PAL through the system/ technology	3	5
	Enhancing the quality of PAL through system feedback and peer feedback	10	0

(1) Enhancing learning effectiveness

Students in the experimental group reported that they could utilize AI feedback (42 times) and teacher feedback facilitated by AI (1 time) to grasp the technical skills. To be more specific, AI helped them identify their weaknesses, establish areas for improvement, and highlight learning issues that prompted them to seek assistance from their teacher.

The system helps me identify my mistakes more accurately. It allows me to focus on those areas for improvement, which enhanced the precision of my poses. I can confidently say that I am indeed more diligent than before. (E1010229)

I have always felt that my thighs are positioned perpendicular to my calves; however, the system consistently indicates otherwise. When the system displays the skeleton diagram, I realize that they are not at a 90-degree angle. This discrepancy requires constant adjustment, and the system helps me understand that I am not performing correctly while guiding me on areas for improvement. (E1060541)

With this learning mode, the system identifies your mistakes. If you still have questions after reviewing the feedback, you can seek further clarification from the teacher. (E1050217)

Students in the control group indicated that they could utilize technological aids (12 times) and feedback provided by the teacher (31 times) to master the technical skills. The photo function on their phones could enhance memory, facilitate review of the learning process, and support self-reflection. Additionally, uploading practice photos of their poses and receiving teacher comments and scores were also key factors that helped students make progress.

With the photos I took on my phone, I can check whether I performed the poses correctly and then show them to the teacher for confirmation or correction. This also allows me to review on my own. (C1020247)

Since I had to upload to the learning platform this time, I put in extra effort, adjusting my angles and becoming more familiar with the poses. I also reviewed the teacher's comments online, which helped me understand where I needed to improve to perform better. (C1040526)

The primary difference between the two groups was that the experimental group exhibited more active learning behaviors and had less need for guidance from the teacher. In contrast, while the control group also reported various learning advantages facilitated by technology, they remained highly dependent on the feedback and guidance of the teacher. This result suggests that AI can facilitate active learning, which not only improves students' learning outcomes, but also reduces teachers' workload.

(2) Improving the quality of PAL

Students in the experimental group (3 times) and the control group (5 times) indicated that the learning system enhanced peer interaction, enjoyment, friendship, and the quality of PAL.

Because someone is taking photos for me, I am motivated to perform better. (C1030410)

It has increased peer interaction as they assist with the photography. (E1030230)

It is more interesting. I can take photos with classmates, which makes it less boring than previous courses. (E1050241).

While taking photos, the person behind the camera also assists in correcting my poses. (C1010227).

The difference between the two groups was that feedback from the system and peer feedback complemented each other, thereby enhancing the quality of PAL (10 times). For instance, there were occasions when the AI did not identify specific areas for improvement; with peer tutoring, students successfully adjusted their poses to the correct form. Afterwards, the AI detection awarded a perfect score, reflecting the positive changes in the students' poses. Conversely, there were times when peers failed to recognize issues. However, through the data and feedback provided by the AI, students became aware of the problems and were able to effectively engage in peer tutoring. These descriptions not only illustrate that students were able to receive personalized guidance from the AI system, but also show that they were able to actively apply the information to adjust their subsequent learning strategies.

I tend to hunch my back during certain poses, but the AI sometimes fails to detect this issue. My peers help correct my pose, which allows me to improve further and achieve a perfect score. (E1010734)

It is a combination of guidance from my group members and the system. Sometimes, my group members may not notice problems, but the system can pinpoint them, helping me improve and achieve better scores. (E1010644)

While my group members may not always be professionals in pose analysis, the data provided by the AI and the program enables them to identify weaknesses, allowing them to adjust my poses. (E1060323)

6 Discussion and conclusion

This study proposed the AI-PAL mode from the perspective of SDT. An AI-facilitated PAL system was developed based on the proposed approach and the test indicators provided by domain experts to monitor students' learning performance and provide adaptive feedback for performance analysis, serving as scaffolding to support students while executing PAL tasks.

In terms of the first research question, the proposed learning mode significantly enhanced students' performance of motor skills, verifying that instructional activities designed based on SDT can effectively support students' learning performance in physical education (White et al., 2021). This result aligns with Völlinger et al. (2023), revealing that integrating SDT and PAL can provide substantial benefits for students' learning performance. Moreover, it is consistent with previous research, indicating that ITS can help students understand their current learning status and provide adaptive feedback to support PAL activities, positively affecting students' learning performance (Han et al., 2021; Haq et al., 2020; Walker et al., 2014).

As for the second research question, the AI-PAL mode significantly enhanced students' overall learning motivation and intrinsic motivation, while no significant difference was found between the extrinsic motivation of the two groups. This result is consistent with the findings of Liu et al. (2025), which revealed that AI-supported PAL activities did not have a significant effect on extrinsic motivation, but autonomy and accessibility provided by AI was effective in terms of enhancing students' intrinsic motivation. Based on the theory of SDT, extrinsic motivation is usually related to rewards and punishments, which are driven by extrinsic factors (Kashefian-Naeeni et al., 2024; Ryan & Deci, 2020). The AI-PAL mode developed in this study was not designed to enhance the mechanisms associated with extrinsic motivation; thus, the results support the core tenets of SDT, demonstrating that learning activities that address students' basic psychological needs can stimulate their intrinsic motivation (Ryan & Deci, 2020). This is also consistent with previous research that explored the impacts of PAL on student learning from the perspective of SDT (Völlinger et al., 2023). On the other hand, past research suggests that students' motivation to engage in learning is significantly increased when they realize that AI can provide them with more effective supports to achieve the learning goals than traditional teaching (Chanda et al., 2025). Previous studies have also indicated that applying human activity analysis to provide feedback on students' skills performance can enhance their learning motivation (Bergner et al., 2021). This implies that the ITS developed

in this study, utilizing OpenPose as the recognition technology, also contributed to improving students' learning motivation.

Regarding the third research question, the AI-PAL mode significantly increased students' basic psychological needs, including autonomy, competence, and relatedness. Through interview analysis, previous research has indicated that the ITS can provide learning environments that foster student autonomy, enabling them to take control of their learning process. The system can alleviate students' fear of making mistakes, motivating them to practice and build confidence; as students complete learning tasks, their competence is reinforced. When students encounter difficulties while interacting with the ITS, consulting with the instructor can enhance their relatedness (Annamalai et al., 2023; Chen et al., 2023). In the present study, the basic psychological needs of both groups were measured using a questionnaire. From a quantitative research perspective, the results align with the aforementioned findings, confirming that learning through the ITS helps fulfill students' basic psychological needs. Also, it echoes Teraoka et al. (2020), noting that providing students with choices during the learning process, such as selecting course content and determining the time spent on learning tasks, can comprehensively satisfy students' three basic psychological needs. In addition, the information presented on the learning analytics dashboard facilitated student monitoring and reflection, supporting self-regulated learning (SRL) (Paulsen & Lindsay, 2024). Specifically, the visual display of learning progress provided by the AI-PAL mode enabled students to monitor their own learning progress, reflect on specific deficits that need to be improved, and adjust their subsequent learning actions. From a theoretical point of view, this design not only helped to motivate students to engage in SRL and strengthen their awareness of independent learning, but also facilitated their interaction with peers in a more meaningful way, thus fulfilling the basic psychological needs in SDT.

In terms of the fourth research question, the AI-PAL model successfully improved the quality of PAL and reduced teachers' workload. Previous research has revealed that some students struggle to understand the feedback provided by the ITS and must depend on instructors for guidance on corrections (Kang et al., 2023; Lin et al., 2023; Sardari et al., 2023). This study incorporated PAL and demonstrated a complementary relationship between system feedback and peer feedback based on student interviews. Students were able to make subsequent practice plans based on peers' feedback and the performance analysis data provided by AI. This finding aligns with Banihashem et al. (2024), showing that providing both AI feedback and peer feedback can create a synergistic effect, supporting students in successfully completing learning tasks. This study further found that such support mechanisms enabled students to rely more on peer and system-generated feedback, thereby reducing the frequency with which they needed to seek direct teacher guidance. Such support suggests the potential to reduce teacher workload while promoting greater student autonomy in the learning process. This shift helped improve the passive learning process whereby students waited for the teacher to input learning information; instead, they actively engaged with their peers and the learning system to comprehend the learning content. Such a learning process and its advantages are also supported by social constructivism (Vygotsky, 1978).

6.1 Theoretical implications

The results of this study validated the applicability of the AI-PAL model based on the Self-Determination Theory (SDT) in physical education. The experimental results showed that the approach significantly increased students' motor skill performance, motivation, and basic psychological needs (including autonomy, competence, and belonging). These results support SDT's claim that when students' basic psychological needs are met, their motivation to learn will increase, which in turn will lead to the realization of their learning goals. Unlike the interview analysis used in previous studies, this quantitative analysis measures changes in students' basic psychological needs to verify that the ITS learning environment is effective in terms of meeting students' psychological needs. These findings not only highlight the close relationship between learning performance, learning motivation and basic psychological needs, but also provide a solid theoretical basis for teachers to utilize AI-assisted learning.

In addition, the AI system developed in this study provides indicators and real-time assessment, so that students can learn to evaluate from an objective perspective by referring to the scoring criteria and examples provided by the teachers. At the same time, through the PAL strategy, students were guided to communicate and discuss with each other, and to compare, reflect, and think critically about the systematic assessment. From the perspective of social constructivism, students will be able to acquire the required knowledge and skills through interaction with peers and learning resources (systems). This will provide strong support and feasible insights for educational institutions to implement AI-assisted learning.

6.2 Practical implications

The present study demonstrated the system development, application, and learning benefits of the AI-PAL mode in physical education. Such findings could extend researchers' and educators' understanding of the application of AI to facilitate PAL, particularly in the context of borrowing human activity analysis (HAA) techniques to enhance the quality of students' verbal and nonverbal PAL in face-to-face communication situations. It involves the ability of AI to effectively support students in a timely manner in learning to objectively assess their peers and provide effective feedback guidance, which is practiced in face-to-face postural adjustments. This finding is critical for physical education, which emphasizes hands-on skills. At the same time, this is a departure from previous research focusing on AI for motor skills. Instead of examining students' individual learning, this study provides a new perspective on PAL activities in physical education. It demonstrates how AI can be used not only to assess individual motor skills, but also as a scaffold for PAL to support peer interaction in physical education. Therefore, the findings of the present study can serve as a reference for not only AI in physical education, but also other skill-related fields such as drama performance and gesture training.

On the other hand, special attention should be paid to the issue of cognitive offloading when applying AI-assisted learning to students, especially when students rely only on the feedback generated by AI (Lahza et al., 2025). In this study, in order to ensure that students used AI-assisted learning correctly and appropriately, we explic-

itly informed students before the beginning of the study. We explicitly informed students at the outset that the test indicators on the system came from expert teachers, that students could use them as scripting scaffolds for the assessment, and that the AI test results provided by the system were to be viewed from an objective point of view. We encouraged students to fully discuss the complementarities and conflicts between the system feedback, peer feedback, and the content of the materials during the PAL process to avoid premature cognitive offloading.

6.3 Limitations and future research

This study has some limitations. First, the teaching experiment was conducted in an authentic teaching environment that aligned with the formal curriculum, and the students were not assigned randomly to groups. Therefore, the research findings should be interpreted with caution. Second, this study analyzed student learning performance before and after the intervention. Considering that SDT-based instructional strategies have the potential to promote students' long-term engagement in physical activities (White et al., 2021), it is suggested that future research adopt true experimental designs to investigate more precise experimental effects. Additionally, future studies could incorporate panel data to understand the long-term impact of SDT on students' learning in physical education. Finally, the survey field of this study is Taiwan's higher education institutions. In Taiwan's higher education institutions, the campuses have high wireless network coverage, and it is also very common for students to have their own smartphones with camera functions. In light of the global gap in student digital literacy and the impact of technology adoption on learning outcomes (Liu & Liao, 2025), the results of this study are more applicable to the study of students' digital literacy in Taiwan and in other countries with similar backgrounds. On the other hand, the effectiveness and feasibility of this AI-assisted learning strategy in other countries with different levels of digital literacy and cultural backgrounds may be a direction for future research.

Authors' contributions All authors contributed to the study conception and design. Methodology, Formal analysis, Investigation, Data Curation, Original Draft and Writing were performed by Lu-Ho Hsia. Formal analysis, Validation, Writing, Review and Editing were performed by Yen-Nan Lin. Technical support, System development, Original Draft, Writing, Review and Editing were performed Chung-Hisneh Lin. Conceptualization, Methodology, Writing, Review and Editing were performed Gwo-Jen Hwang. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The data and materials are available upon request to the corresponding author.

Code availability Not applicable.

Declarations

Ethical approval The ethical requirements for research in this selected university were followed.

Consent to participate The participants all agreed to take part in this study.

Consent for publication The publication of this study has been approved by all authors.

Conflicts of interest There is no potential conflict of interest in this study.

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